

Ayush Pondugula

678-643-7325 | apondugula3@gatech.edu | linkedin.com/in/ayushpondugula/ | github.com/ayushp211 | US Citizen

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, GPA: 3.71/4.0

May 2028 (Expected)

- Concentrations in **Artificial Intelligence** and **Systems & Architecture**
- Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Intro to Artificial Intelligence, Machine Learning, Systems & Networks, Programming for HW/SW Systems

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Flock Safety

Atlanta, GA

- Engineered a concurrent data pipeline (**ClickHouse**, **AWS S3**) synchronizing **2,000+ internal images** across **150+ cameras** with 360° Google Street View panoramas, generated by stitching 512 map tiles per endpoint.
- Architected an algorithmic **model harness (DSPy)** cross-analyzing LPR images and 360° panoramas via landmark extraction and shadow trajectory heuristics to deduce spatial headings, eliminating manual calibration.
- Developed a robust validation suite targeting an **85%+ accuracy** threshold to flag low-confidence camera data.
- Deployed a **React** mapping UI to visualize dynamic FoV cones across **150+ cameras**, expediting investigations.

Undergraduate Research Engineer

Aug. 2025 – Present

Georgia Institute of Technology VIP: AI Makerspace Nexus

Atlanta, GA

Industry Partners: Nvidia, GT College of Engineering, PACE Team, OLIVES Lab

- Democratized access to **HPC** resources, enabling **80+ non-technical students** to leverage GenAI resources.
- Managed PACE-ICE enterprise **NVIDIA H100/H200 GPU** clusters, ensuring **99.9% service availability**.
- Led an **Agile team of 5** to deliver **1 major feature per sprint**, accelerating the project timeline by **2 weeks**.

PROJECTS

BuzzPlayer | Python, React, Flask, vLLM, Whisper, NumPy, SentenceTransformers

Jan. 2026 – May 2026

- Designed a full-stack chatbot powered by a **multi-modal RAG** pipeline for **timestamp-grounded video QA**.
- Fused audio-visual streams via **temporal chunking**, implementing **sub-second cosine similarity search**.
- Partitioned an **8-GPU** cluster to serve concurrent **106B-parameter** LLMs using vLLM and Tensor Parallelism.
- Enforced strict XML boundaries and **greedy decoding** to eliminate hallucinations and improve output accuracy.

Study Buddy | FastAPI, React, LangChain, vLLM, Docker/Singularity, PyMuPDF

Aug. 2025 – Dec. 2025

- Developed a **local RAG** for personalized quizzes, achieving **95% accuracy** and **100% HPC data sovereignty**.
- Engineered a preloaded document workflow **reducing time-to-generation by 70% (Approx. 45s to 13s)**.
- Implemented dynamic CORS handling via FastAPI and Regex for rotating cluster IPs, ensuring **zero downtime**.

Cave Explorer Game Engine | C++, ARM Mbed

Oct. 2025 – Dec. 2025

- Architected a custom C++ game engine to orchestrate rendering, inputs, and collision for real-time gameplay.
- Built a custom Hash Table via **manual dynamic allocation** for sparse maps, **reducing memory use by 40%**.
- Engineered **FSM** logic to handle **9 interactive entities** (incl. multi-stage Boss AI) with **0 performance drops**.

Event Ticketer | Python, Django, HTML/CSS, JavaScript, SQLite

Oct. 2025 – Dec. 2025

- Architected a secure Django marketplace with custom **RBAC** to enforce permissions across **3 user roles**.
- Developed a real-time chat service using **AJAX polling**, **reducing page reloads by 80%** to optimize UX.
- Processed orders via **atomic transactions**, ensuring **zero inventory discrepancies** for a live SQL dashboard.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, Assembly (RISC-V), JavaScript, TypeScript, HTML/CSS

Frameworks: FastAPI, Flask, Django, React, Tailwind CSS, JUnit, DSPy

Tools: Git, AWS (S3, Bedrock), ClickHouse, Docker/Singularity, VSCode, Linux/Unix CLI, Jira, ARM Mbed, LaTeX

Libraries: vLLM, LangChain, Transformers, NumPy, JAX, Pandas, SentenceTransformers, PyMuPDF